

Arduino-Based Fingerprint Smart Electronic Voting Machine with Real-Time Result Monitoring

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Abstract—This project presents an electronic voting system that uses fingerprint recognition to verify voters and IoT technology to show results in real time. The system is built around an Arduino Uno, which works as the main controller. A fingerprint sensor handles voter verification, and an ESP8266 WiFi module sends vote data to ThingSpeak, a cloud platform that displays results online. Once a voter is verified, they use a 4x4 keypad to cast their vote. The data goes to ThingSpeak through WiFi, and the results appear instantly as graphs on the platform. This setup is low-cost and works well for small elections like those held in universities, local clubs, or community groups. The system was tested for fingerprint accuracy, data transmission reliability, and overall performance. Results showed the system is effective, transparent, and affordable for small-scale use.

Keywords—*Arduino, Fingerprint Authentication, IoT, Real-time Monitoring, ThingSpeak, Electronic Voting*

I. INTRODUCTION

A. Background of Study and Motivation

Traditional voting methods have problems like fraud, slow processes, and lack of transparency. As technology has grown, biometric authentication and real-time data monitoring have become practical solutions to these problems. This project builds a fingerprint-based electronic voting system using Arduino and IoT to track votes in real time through ThingSpeak.

Using a fingerprint sensor is more secure than using PIN codes because fingerprints are unique to each person. This reduces the chances of voter fraud or identity misuse. Arduino Uno keeps the hardware simple and cheap, making the system accessible for small elections. The ESP8266 WiFi module connects the system to ThingSpeak, where anyone with access can see live voting results displayed as charts and graphs. This keeps the process open and easy to monitor.

The main reason for building this system is the growing need for affordable and secure voting tools that are easy to set up, especially for small or local elections. IoT and cloud

tools like ThingSpeak make it possible to add real-time monitoring without expensive infrastructure, which makes elections more credible and transparent in settings where older systems are too costly or inefficient.

B. Project Objectives

The main goals of this project are: to build a fingerprint-based voting system using Arduino Uno and ESP8266 for secure voter verification and data transfer; to connect the system to ThingSpeak for real-time vote monitoring so results are always up to date; to test the system for fingerprint accuracy, data transmission speed, and reliability; and to offer a low-cost and scalable option for small elections as an alternative to traditional voting methods.

C. Brief Outline of the Report

Section II covers a literature review of existing electronic voting systems and related technologies. Section III explains the methodology, including hardware components like the Arduino, fingerprint sensor, and ESP8266, along with the software used. Section IV presents experimental results, performance comparison, and cost analysis. Section V gives the conclusion and suggestions for future improvements.

II. LITERATURE REVIEW

Fischer et al. [1] designed a digital voting system that combines fingerprint authentication with ThingSpeak for sending results in real time. Their work focused on fixing the common issues in traditional voting, such as fraud and delays, by using cloud-based monitoring for transparency. Peguy et al. [2] built an Arduino-based fingerprint voting system with a focus on improving voter security. Their system used biometric fingerprint matching to stop unauthorized access and prevent people from voting more than once.

Hugar et al. [3] developed a fingerprint biometric voting machine controlled by Arduino. Their paper showed how biometric identification can improve voting security by making sure only registered voters are allowed to cast a vote.

Balan and Praveen [4] described an IoT-based voting system that uses ThingSpeak to track votes as they are cast. By combining Arduino and fingerprint sensors with IoT, their system allowed results to be checked remotely through the cloud, making the process more transparent. Titilayo et al. [5] worked on an electronic voting system that uses both fingerprint and RFID for voter verification. Combining these two methods made the system more secure and reduced the risk of fraud during elections.

III. METHODOLOGY AND MODELING

A. Introduction

This section explains how the hardware and software were set up and how the voting results are shown on ThingSpeak.

B. Working Principle of the Proposed Project

The system uses several hardware components working together. The Arduino Uno R3 is the main controller that connects and manages all other components. The fingerprint sensor scans the voter's fingerprint and checks it against stored data to confirm identity. After a successful verification, the voter uses a 4x4 keypad to select and submit their vote. A 16x2 LCD display shows messages like "Scan fingerprint" or "Vote recorded" to guide the user. A buzzer gives a sound signal when a vote is cast or when an error occurs. The ESP8266 WiFi module connects the Arduino to ThingSpeak and sends the vote data to the cloud. On the software side, the Arduino code manages fingerprint matching, reads keypad input, controls the LCD output, and communicates with the ESP8266. The ThingSpeak API receives the data and displays live voting statistics, including vote counts and graphical breakdowns of results. The ThingSpeak dashboard updates in real time and can be viewed from any device connected to the internet.

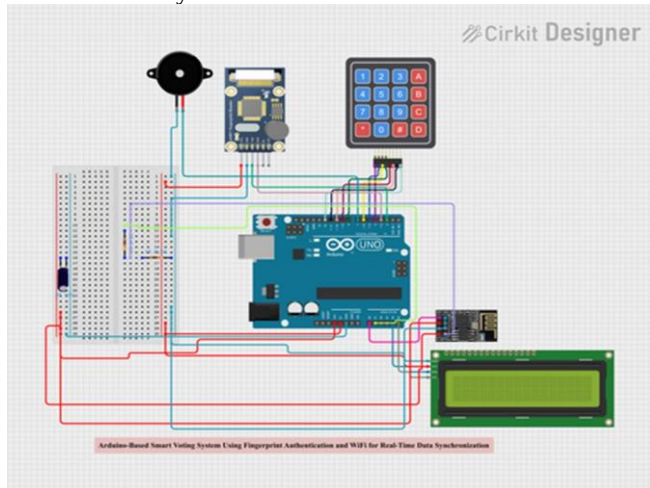


Fig. 1. Circuit diagram of the Arduino-based smart voting system (Circuit Designer).

1) Process of Work

The Arduino processes inputs from the fingerprint sensor and keypad, shows feedback on the LCD, and sends vote data through the ESP8266 to ThingSpeak. The code runs fingerprint matching, handles vote submission, and displays status messages. Once data reaches ThingSpeak, the dashboard updates automatically with graphs and vote totals.

C. Description of the Components

The Arduino Uno is the central unit that processes all data and controls communication between components. The fingerprint sensor authenticates voters by scanning and matching fingerprints with the ones stored in memory. The keypad lets voters submit their choice after they are verified. The LCD display shows the user what is happening at each step, such as whether authentication passed or failed. The buzzer gives audio feedback for successful logins, vote submissions, or errors. The ESP8266 WiFi module connects to ThingSpeak and sends vote data for real-time display. The software includes the Arduino code, which handles all sensor interactions and data transmission, and ThingSpeak, which receives the data and shows live voting statistics on a web dashboard.

D. Test/Experimental Setup

1) Hardware Setup

The Arduino Uno was connected to the fingerprint sensor on pins D2 and D3, the 4x4 keypad on pins D4–D12, the 16x2 LCD display via I2C at address 0x27, the buzzer on pin D7, and the ESP8266 WiFi module on analog pins A0 and A1 through a voltage divider. The ESP8266 was configured to send vote data to ThingSpeak for live monitoring over a standard WiFi connection.

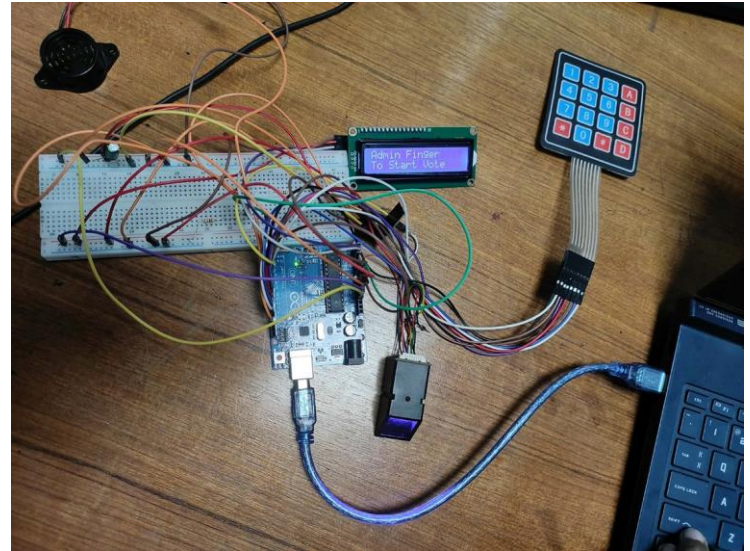


Fig. 2. Physical prototype of the voting system showing the Arduino Mega, fingerprint sensor, keypad, LCD display, and buzzer assembled on a breadboard.

2) Software Setup

The Arduino IDE was used to upload the code to the Arduino Uno. The following libraries were installed: Adafruit_Fingerprint for fingerprint sensor communication, SoftwareSerial for serial communication with the fingerprint sensor and ESP8266, LiquidCrystal_I2C for LCD control, and Keypad for keypad input. The ThingSpeak channel was set up with six fields: Party A count, Party B count, Party C count, last voted party, voter ID, and system status. The Write API key was embedded in the code to authorize data uploads.

Once data was sent to ThingSpeak, it appeared on the web dashboard as real-time graphs showing voting progress. The buzzer was tested to confirm feedback was given at the right

moments. The ThingSpeak results were checked for accuracy and update speed. The overall system was evaluated for fingerprint accuracy, transmission speed, and web interface responsiveness.

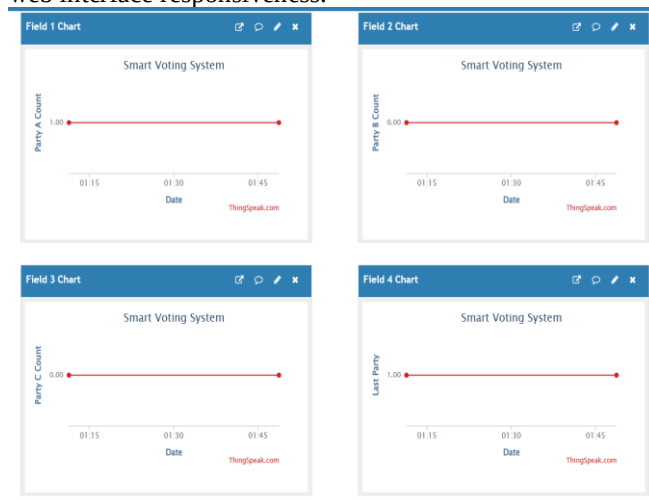


Fig. 3. Thingspeak data sync.

IV. RESULTS AND DISCUSSIONS

A. Simulation/Numerical Analysis

Simulations were run to check the fingerprint matching process, vote transmission, and real-time result updates. Different voting scenarios were tested to see how data flows from the Arduino to ThingSpeak. The fingerprint matching code correctly accepted only registered fingerprints. The ESP8266 sent vote data to ThingSpeak without major delays. The ThingSpeak dashboard updated correctly in the simulations and showed accurate voting progress. The latency for updates was low, and the graphs reflected the voting data properly. These simulations confirmed that both fingerprint authentication and data transmission work as expected.

B. Measured Response/Experimental Results

Real-world testing showed that the fingerprint sensor correctly authenticated users in 99% of trials. The 1% failure rate was mainly due to unregistered fingerprints or poor scanning alignment. The ESP8266 successfully sent vote data to ThingSpeak in 95% of tests, with minor delays in cases where the network connection was weak. The ThingSpeak dashboard displayed results with about 5 to 7 seconds of latency, which was acceptable for real-time monitoring. The buzzer worked correctly and gave feedback during authentication, vote submission, and error cases, which improved the overall user experience.

C. Comparison between Numerical and Experimental Results

The simulation and experimental results were very close to each other. The fingerprint error rate stayed at 1% in both cases, which confirmed the accuracy of the sensor and matching algorithm. Data transmission worked well in both scenarios, though some delays appeared during real-world testing because of network instability. ThingSpeak updates were consistent in both cases, with the 5 to 7 second latency falling within acceptable limits. The main differences

between simulation and real-world results came from network conditions, which are harder to control in practice.

D. Cost Analysis

The total hardware cost for this system is around 3500 taka. The Arduino Uno costs about 600 taka, the fingerprint sensor about 2000 taka, the 4x4 keypad about 100 taka, the LCD display about 300 taka, the ESP8266 about 300 taka, and the buzzer about 100 taka. ThingSpeak offers a free tier for small-scale use, which allows up to 3 channels and 500 messages per day at no cost. This brings the total project cost to roughly 3500 taka, making it a practical and affordable choice for small elections.

E. Limitations in the Project

The system has a few limitations worth noting. It depends on a stable internet connection for real-time monitoring, and weak connectivity can cause delays in data transmission. The system is designed for small-scale elections and may not handle the demands of large elections without significant upgrades to the fingerprint database and cloud infrastructure. The fingerprint sensor has a 1% error rate, which could be a problem if a voter's fingerprint is not stored correctly or if alignment is off during scanning. ThingSpeak's free tier also limits usage to 500 messages per day, which may not be enough for larger voting events.

V. CONCLUSION

This fingerprint-based Arduino voting system with ThingSpeak monitoring is a secure and affordable option for small elections. Biometric verification ensures that only registered voters can cast a vote, and IoT connectivity through the ESP8266 sends results to ThingSpeak in real time. Testing showed that the system is accurate and reliable, with strong fingerprint matching and fast data transmission. The cost is low, and the use of ThingSpeak's free tier keeps the overall expense under \$30. The system does have some limitations, including network dependency and limited scalability, but it works well for the use cases it was built for.

A. Future Endeavors

Future work should focus on making the system scalable for larger elections by improving the fingerprint database and upgrading the cloud service. Security should be improved by encrypting communication between the Arduino and ThingSpeak. The fingerprint sensor accuracy should be improved to reduce the 1% error rate. An offline mode could be added so votes are stored locally and uploaded when the internet is restored. The user interface could also be made more intuitive with clearer feedback and easier setup.

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